

PATHFINDER

Discover Greece's Hidden Trails

AI-Powered Sustainable Outdoor
Discovery for Greece.

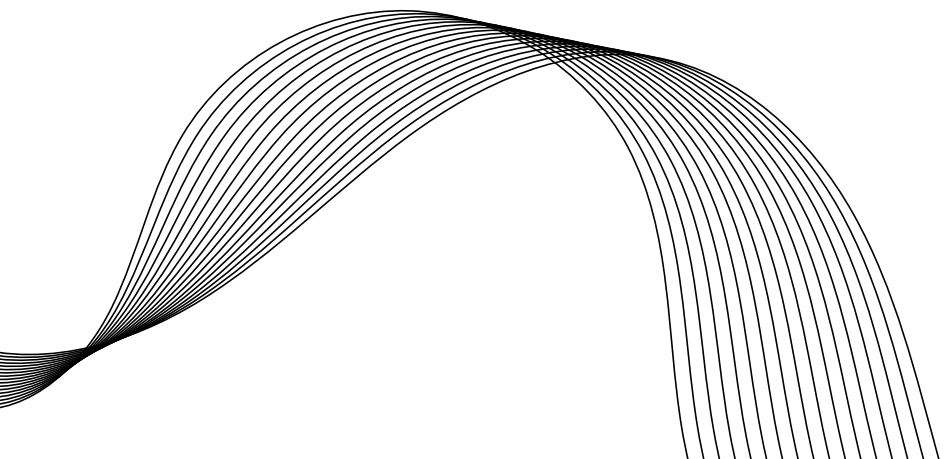
Built for the Deloitte Makeathon
2026 - Sustainable Tourism
Challenge.

Explore Now



THE PROBLEM: UNEQUAL DISTRIBUTION

2.4 million tourists flooded Santorini last year. Meanwhile, 80% of Greece's pristine trail assets saw fewer than 500 visitors.



OUR SOLUTION:

Greece's first AI trail companion built to fight overtourism.

EXPLORE

REDISTRIBUTE

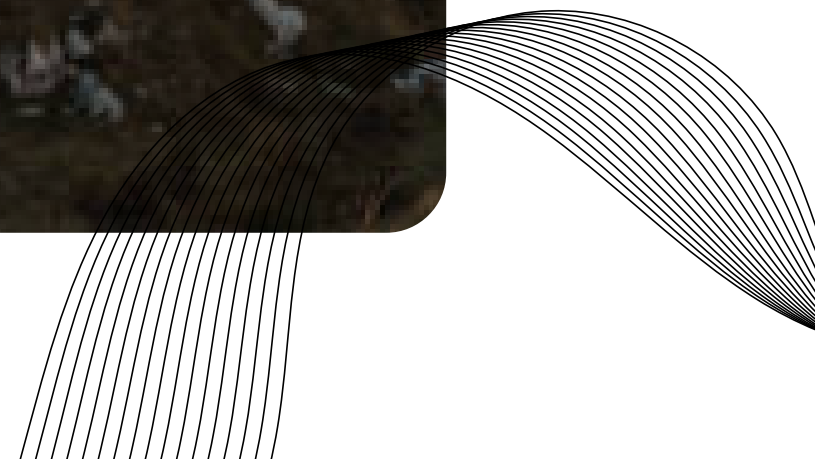
CONNECT

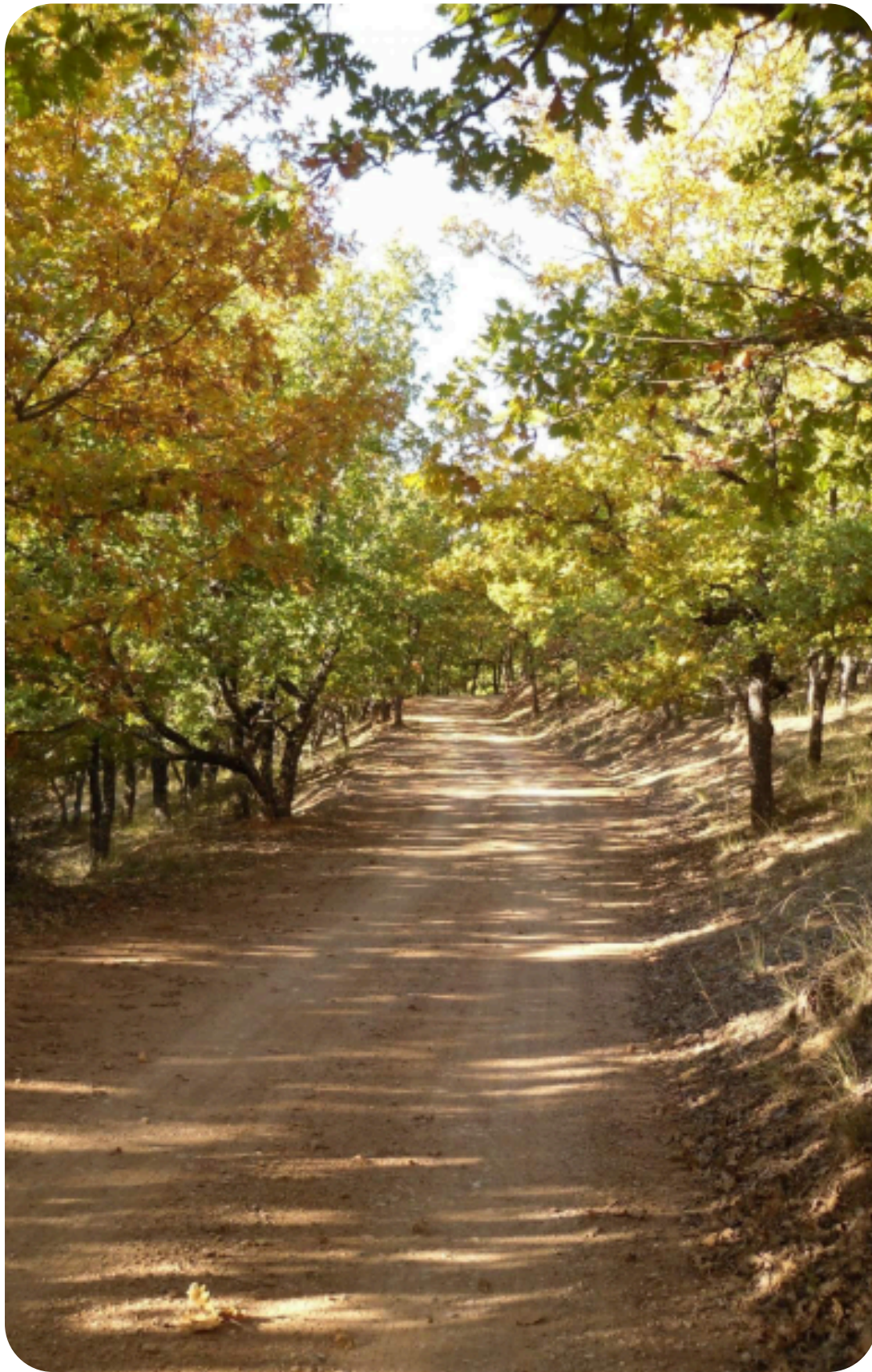
An intelligent platform that actively steers hikers away from saturated spots and toward Greece's 1K+ undervisited trails Combines real-time environmental data with AI personalization Designed for both tourists discovering Greece and Greeks rediscovering their own country



KEY FEATURES

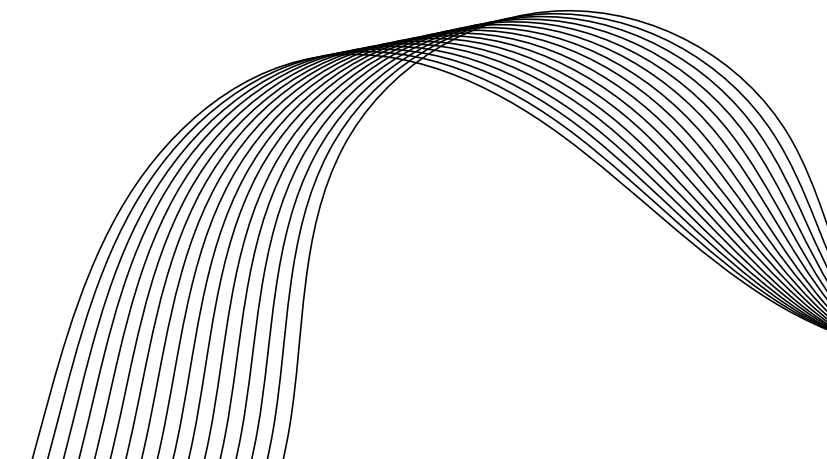
- Personalised Trail Recommendation
- Discovery & Dynamic Profiling
- Planner & Itinerary Generator
- Weather - Events - POIs
- Trail Info & Chatbot Assistance
- Collaborative AI Rerouting
- Trail Matcher Based on Mood
- Community Groupchats





THE DESIGN

- Conversational AI Intake
- Core Geo-Data Foundation
- Live Multi-API Enrichment
- Algorithmic Sustainability Filter
- Context-Aware Narrative Generation



TECHNICAL DETAILS



Architecture (high-level): Client → FastAPI REST API → domain modules (enrichment, routing, weather) → external services (Weather, iNaturalist, Bedrock); stateless API, isolated modules for testability.

Backend: Python + FastAPI (api/main.py) - modular services (trail engine, weather, routing, chat). LLM / AI: AWS Bedrock via boto3 for multi-turn trail chat and "surprise" matching.

External APIs: OpenWeatherMap (weather), iNaturalist (biodiversity), POI calls via requests.

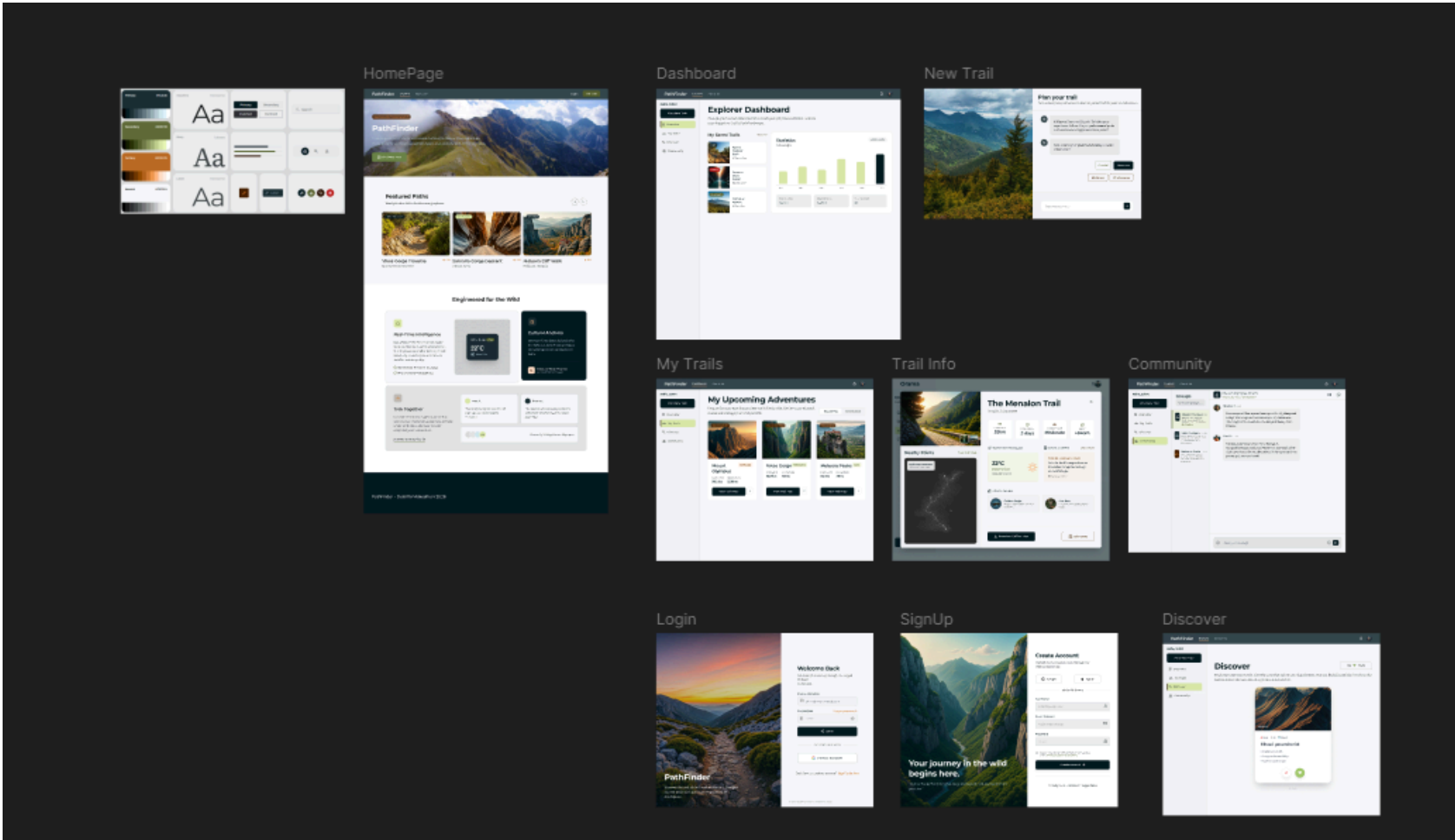
Data: Local JSON store in data (trails.json, users.json); lightweight saved-trails & live-interest persistence.

TECHNICAL DETAILS - FRONTEAD

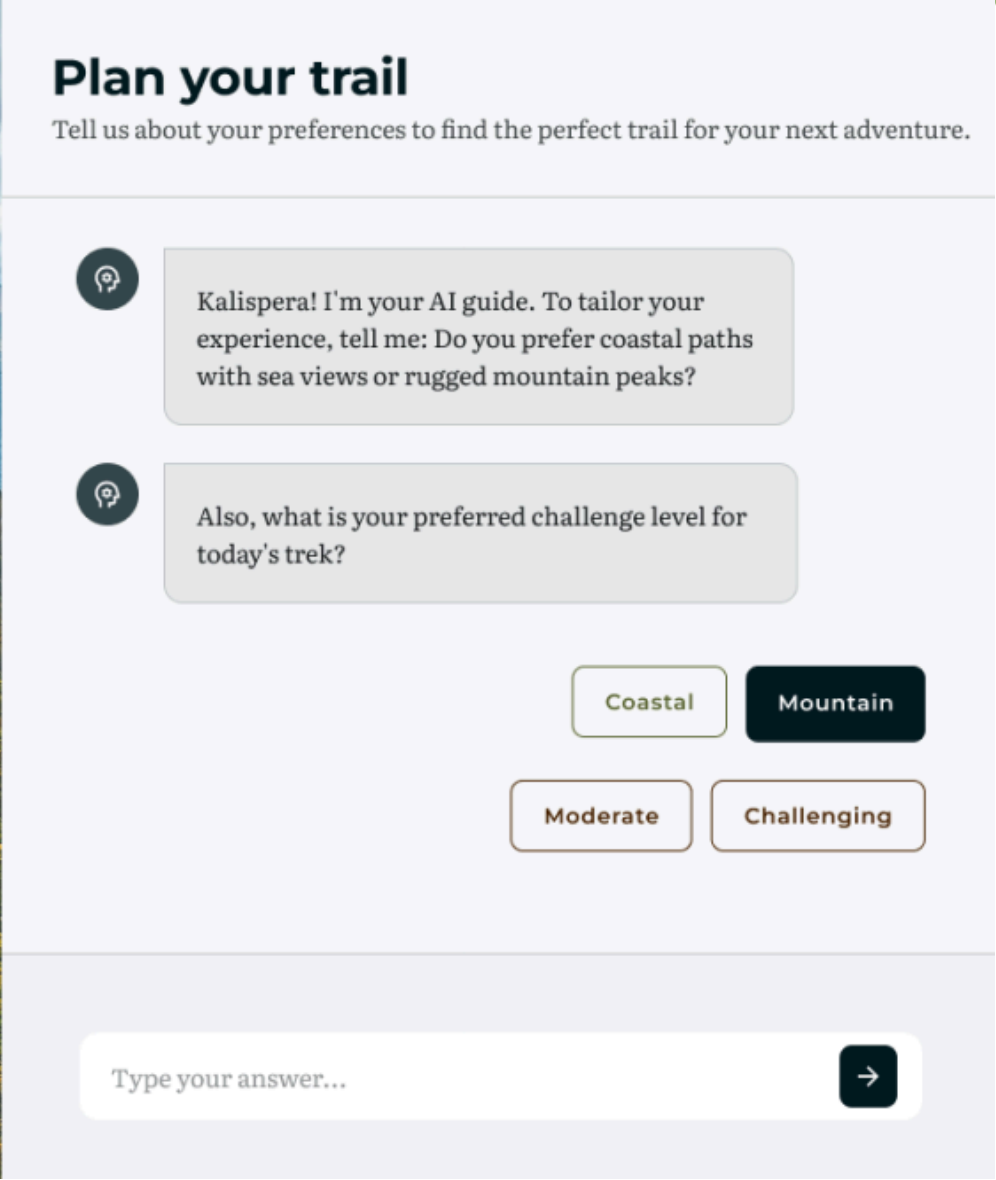
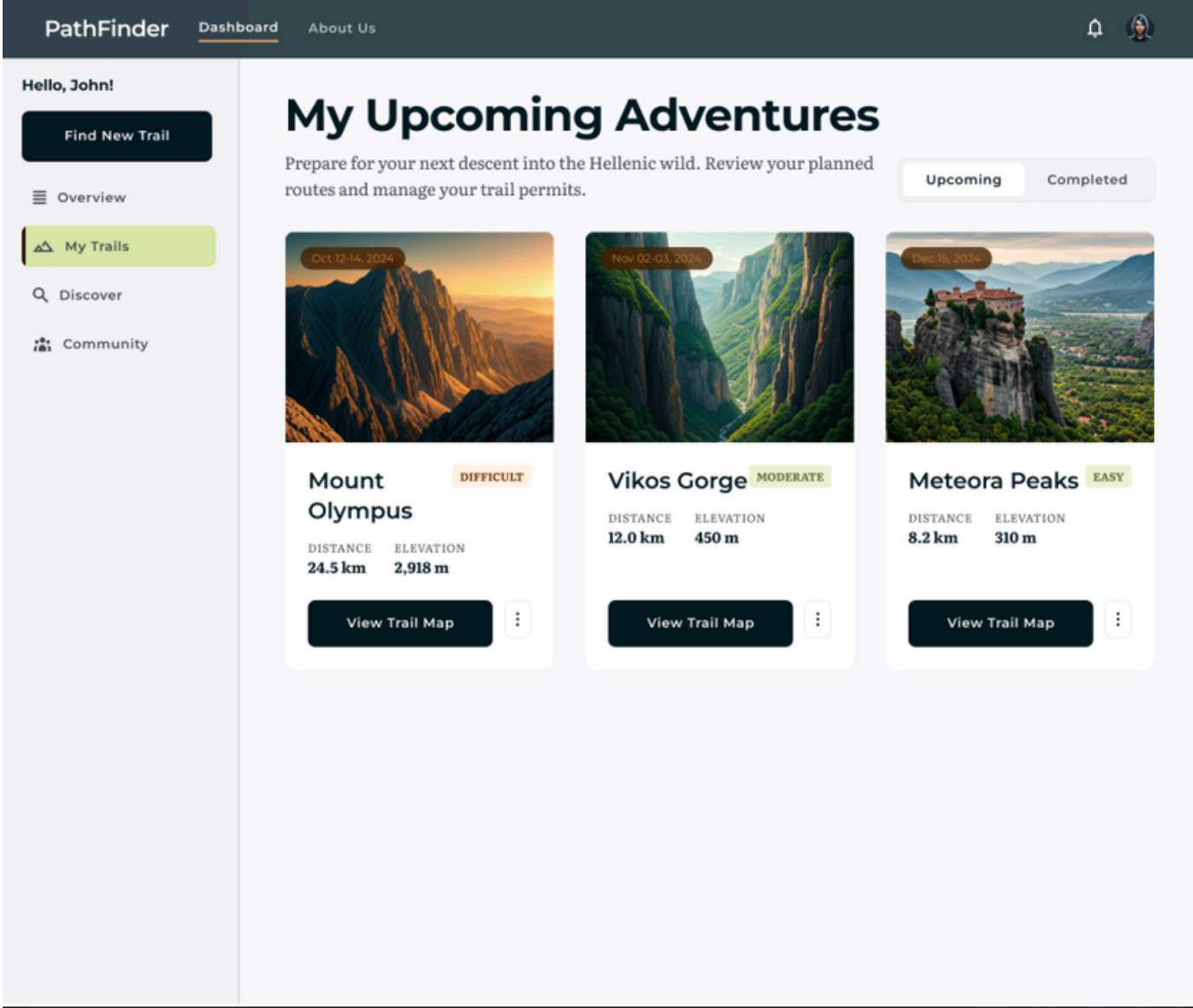
Frontend: React + Vite, axios for REST, leaflet / react-leaflet for maps, react-router-dom for navigation. Design: UI frames & flows prototyped in Figma → implemented as React components.



DESIGNING IN FIGMA



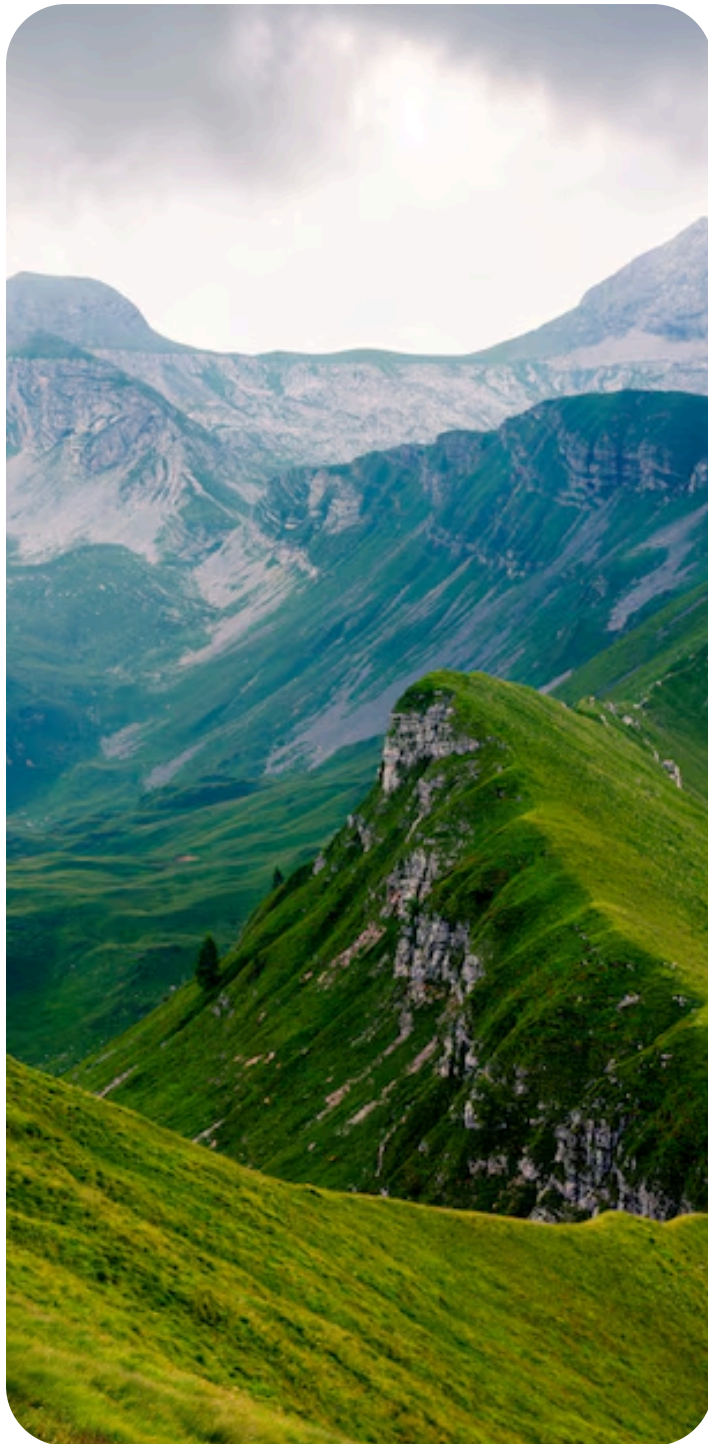
FIGMA





EMPOWERING SUSTAINABLE TOURISM

- Combating Overtourism
- Data-Driven Routing
- Spreading Economic Impact
- Computational Sustainability

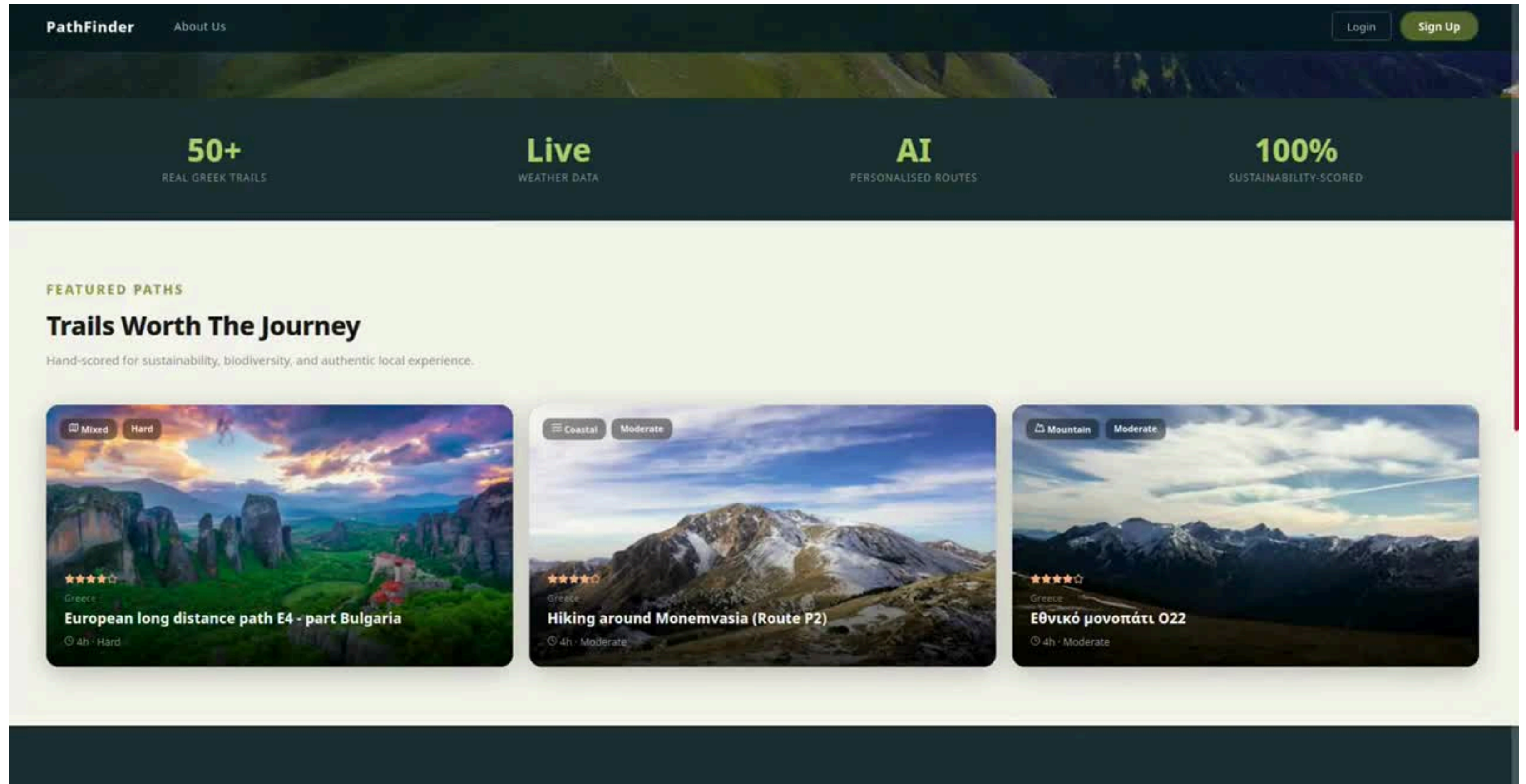


FUTURE PLANS

- Native Mobile Application
- Photo-to-Trail Matching (Visual Search)
- Sustainable Gamification & Local Rewards
- Live Trail Conditions Updates



DEMO





THANK YOU

Despoina Kampioti
Konstantinos Katrakis
Maria Kapaki
Kyriaki Kalamari

